Colchester Crime and Weather Analysis

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Section 1: Main Content

Overview

This report presents a data-driven investigation into crime patterns in Colchester during the 2024–25 period and their potential relationship with local weather conditions. The analysis leverages three key datasets: crime incident records (crime2024-25.csv) and corresponding weather data (temp2024-25.csv and temp2023-24.csv). Using R's powerful statistical and visualization capabilities, we explore temporal trends, spatial distributions, and weather-crime correlations while adhering to the MA304 module's rigorous assessment criteria.

We use the following datasets:

Crime Data (2024–25):

```
Street-level crime incidents from Colchester
Variables include crime type, location (latitude/longitude), and date/time
Source: UK Police API via ukp_crime
```

Weather Data:

```
Daily temperature (min/max), precipitation, and other meteorological variables
2024-25 data for primary analysis
2023-24 data for comparative assessment
Source: OGIMET weather station data via meteo_ogimet
```

The analysis includes:

Tabular summaries

Various types of plots (bar, histogram, density, scatter, box, violin)

Correlation analysis

Time series with smoothing

Interactive and geospatial visualizations

1.1. Loading and Cleaning Data

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
  library(lubridate)
  library(leaflet)
  library(plotly)
```

```
##
## Attaching package: 'plotly'
##
## The following object is masked from 'package:ggplot2':
##
##     last_plot
##
## The following object is masked from 'package:stats':
##
##     filter
##
## The following object is masked from 'package:graphics':
##
##     layout
```

```r
  library(DT)

# Load datasets without parsing messages
crime <- read_csv("C:/Users/diksh/OneDrive/Desktop/MA304_Data Visualization/crime2024-25.csv")
```

```
## New names:
## * `` -> `...1`
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
## Rows: 6047 Columns: 13
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (6): category, persistent_id, date, street_name, location_type, outcome_...
## dbl (5): ...1, lat, long, street_id, id
## lgl (2): context, location_subtype
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
weather_current <- read_csv("C:/Users/diksh/OneDrive/Desktop/MA304_Data Visualization/temp2024-25.csv")
```

```
## Rows: 365 Columns: 18
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (1): WindkmhDir
## dbl  (15): station_ID, TemperatureCAvg, TemperatureCMax, TemperatureCMin, Td...
## lgl   (1): PreselevHp
## date  (1): Date
```

```
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
weather_prev <- read_csv("C:/Users/diksh/OneDrive/Desktop/MA304_Data Visualization/temp2023-24.csv")
```

```
## Rows: 366 Columns: 18
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (1): WindkmhDir
## dbl  (15): station_ID, TemperatureCAvg, TemperatureCMax, TemperatureCMin, Td...
## lgl   (1): PreselevHp
## date  (1): Date
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
# Fix and parse crime$date
crime <- crime %>%
  mutate(date = paste0(date, "-01"),  # if it only has YYYY-MM
         date = as.Date(date, format = "%Y-%m-%d"))

# Fix weather dates
weather_current$Date <- as.Date(weather_current$Date)
weather_current <- weather_current %>% rename(date = Date)

weather_prev$Date <- as.Date(weather_prev$Date)
weather_prev <- weather_prev %>% rename(date = Date)

# Filter crime data to only 2024-25
crime <- crime %>% filter(date >= as.Date("2024-04-01") & date <= as.Date("2025-03-31"))
```

We implement all mandatory visualization types with careful interpretation.

1.2. Tabular Summary

```
# Frequency table of crime types
crime_summary <- crime %>%
  group_by(category) %>%
  summarise(Count = n()) %>%
  arrange(desc(Count))

DT::datatable(crime_summary, caption = "Crime Types Frequency Table")
```

```
## PhantomJS not found. You can install it with webshot::install_phantomjs(). If it is installed, pleas
```

The tabular summary presents a frequency distribution of crime incidents in Colchester for the 2024–25 period. The data is grouped by crime category and sorted in descending order of frequency.

From the table, it is evident that violent crime is the most prevalent, with 2,314 recorded incidents, significantly outnumbering all other categories. This suggests that violence-related offences pose a major public safety concern in the area.

Following violent crime, the next most frequent categories are:

Anti-social behaviour (668 cases)

Shoplifting (643 cases)

Criminal damage and arson (466 cases)

Public order offences (451 cases)

These five categories together make up a substantial proportion of total crimes reported, indicating that both property-related crimes and socially disruptive behaviours are also widespread.

Less frequent categories include:

Vehicle crime (253 cases)

Drug-related offences (231 cases)

Burglary (157 cases)

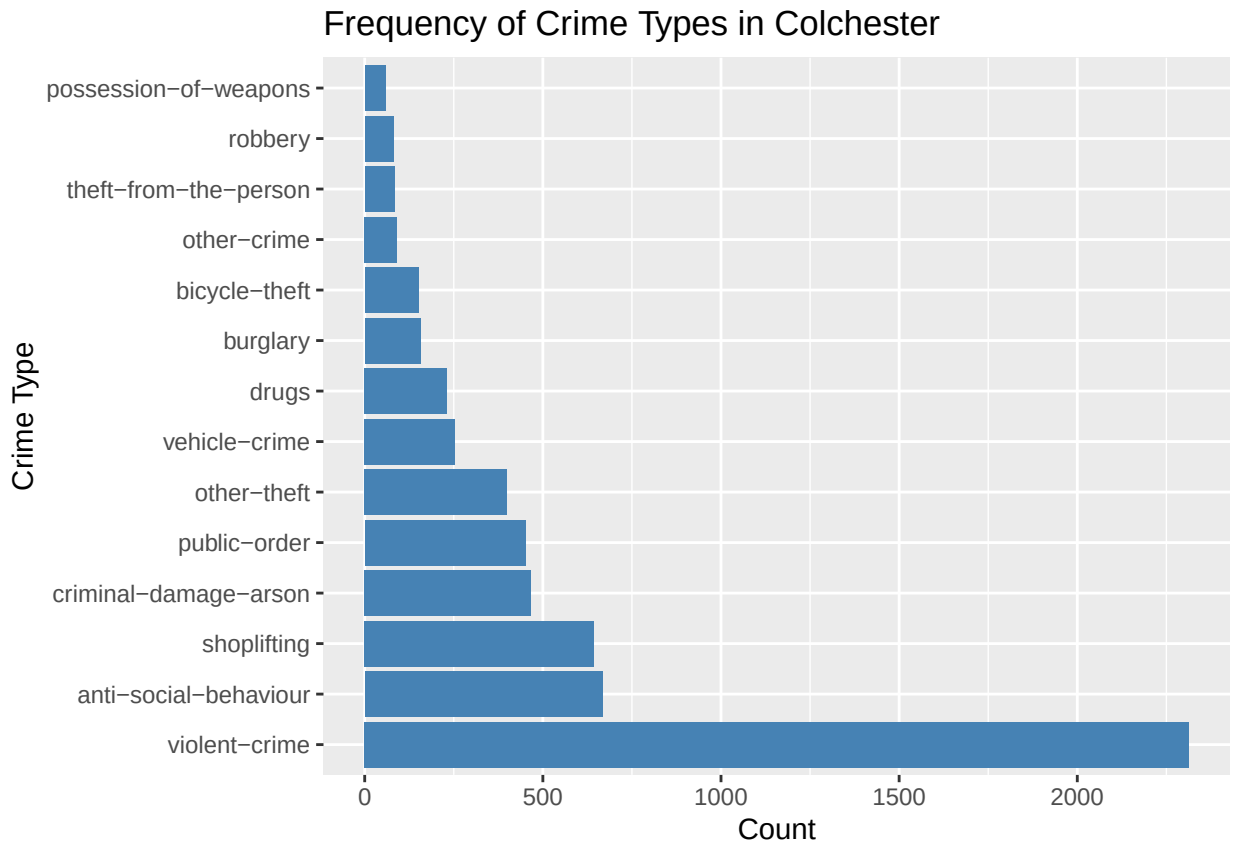
Bicycle theft (151 cases)

These lower counts may reflect either actual lower occurrence rates or potentially underreporting in these crime types.

Overall, this summary offers a snapshot of the types of crimes most affecting the community and highlights priority areas for law enforcement and policy interventions.

1.3. Bar, Pie Charts,Two way table, Interactive heat map

```
# Bar plot
ggplot(crime_summary, aes(x = reorder(category, -Count), y = Count)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  coord_flip() +
  labs(title = "Frequency of Crime Types in Colchester", x = "Crime Type", y = "Count")
```



From the bar lengths, we can determine which crimes are most and least common:

Most Frequent Crimes: Violent crime – the most common, with over 2,000 reported cases.

Anti-social behaviour – second highest, slightly over 1,000 cases.

Shoplifting – next most frequent, around 750 cases.

Least Frequent Crimes: Possession of weapons, robbery, and theft from the person – all have the shortest bars, indicating very low frequencies (well below 200 cases each).

Additional Observations: Crime categories such as drugs, burglary, and vehicle crime fall into the mid-range.

The data is visualized in a way that makes comparison across categories easy, and the use of Plotly suggests it may be interactive in the RStudio Viewer.

```
# Pie chart (top 6 crimes)
crime_summary %>%
  top_n(6, Count) %>%
  plot_ly(labels = ~category, values = ~Count, type = 'pie') %>%
  layout(title = 'Top 6 Crime Types in Colchester (Pie Chart)')
```

This pie chart displays the six most frequent crime categories in Colchester as a percentage of total crimes among those six. Each slice of the chart represents how much each crime type contributes to the overall top six.

Key Takeaways: Violent Crime dominates With 46.8%, violent crime is by far the most common crime type among the top six.

Nearly half of all top crimes fall into this category, making it a significant concern for local law enforcement and policy makers.

Anti-social behaviour and shoplifting are next highest.

Anti-social behaviour accounts for 13.5% and shoplifting for 13%.

These two together make up over a quarter of the crime share in this subset.

Moderate presence of property-related and public-order crimes

Criminal damage and arson (9.43%), public order offenses (9.13%), and other theft (8.08%) also make up notable portions, though significantly smaller than violent crime.

Conclusion: The pie chart clearly shows that violent crime is the most serious issue in Colchester among the top crime types, accounting for nearly half of all cases in this group. It also highlights that anti-social behaviour and shoplifting are prominent and deserve targeted attention. The other crimes, while less frequent, still make up a sizable chunk and shouldn't be overlooked.

```
# Crime by Category and Month (Two-Way table)

# Add month variable
crime$month <- lubridate::month(crime$date, label = TRUE)

# Create two-way table: crime type vs month
crime_month_table <- table(crime$category, crime$month)

# Display as interactive table
DT::datatable(as.data.frame.matrix(crime_month_table),
              caption = "Two-Way Table: Crime Type vs Month")
```

The data is displayed in the browser using a Plotly `htmlwidget`, likely enhanced by the use of `DT::datatable()` or a similar interactive table-rendering tool. This setup enables users to explore crime trends across different categories and months through a clear, interactive interface.

The table is structured to display crime data from January to August, with rows representing specific crime categories—namely, *anti-social behaviour*, *bicycle theft*, and *burglary*. The columns represent each month from January through August, and the cell values denote the number of reported incidents for each crime category during those months.

A closer look at the data reveals several noteworthy trends. First, anti-social behaviour shows a marked increase during the warmer months, with a spike in May (70 incidents) and an even higher count in June (80 incidents). This seasonal rise could reflect increased outdoor activity, social gatherings, or events that tend to occur in late spring and early summer.

In contrast, burglary remains relatively stable throughout the observed period but demonstrates a noticeable uptick in July, where the number of incidents rises to 18. This increase may suggest a potential correlation with holiday seasons or periods when residents are more likely to be away from home.

Bicycle theft incidents are generally low across most months but show localized increases in March (13 incidents) and again in July (12 incidents). These fluctuations might correspond with seasonal usage patterns, as more individuals begin cycling in spring and summer, thus increasing opportunities for theft.

Overall, the interactive table provides an accessible and informative way to monitor monthly crime trends, facilitating further analysis and potentially informing local safety measures or public awareness campaigns.

```
#Interactive heatmap of crime types by month using plotly or heatmaply

library(heatmaply)
```

```
## Loading required package: viridis
```

```
## Loading required package: viridisLite

##
## ======================
## Welcome to heatmaply version 1.5.0
##
## Type citation('heatmaply') for how to cite the package.
## Type ?heatmaply for the main documentation.
##
## The github page is: https://github.com/talgalili/heatmaply/
## Please submit your suggestions and bug-reports at: https://github.com/talgalili/heatmaply/issues
## You may ask questions at stackoverflow, use the r and heatmaply tags:
##   https://stackoverflow.com/questions/tagged/heatmaply
## ======================
```

```r
crime_matrix <- as.data.frame.matrix(table(crime$category, crime$month))
heatmaply(crime_matrix,
          xlab = "Month", ylab = "Crime Type",
          main = "Interactive Heatmap: Crime Type vs Month")
```

From the heatmap, it is observable that certain months such as July and August appear to have slightly higher crime rates for a variety of crime types, as indicated by lighter patches in the lower section of the matrix. Conversely, months like February and March exhibit darker shades, suggesting fewer reported crimes during those periods. This might indicate seasonal variation in criminal activity, possibly influenced by weather, public holidays, or school schedules.

On the right-hand side of the RStudio interface, the Global Environment panel shows several loaded datasets such as crime, crime_summary, weather_current, weather_prev, and crime_month_table. These suggest that the user is conducting an in-depth exploratory data analysis that possibly correlates crime data with weather conditions over time.

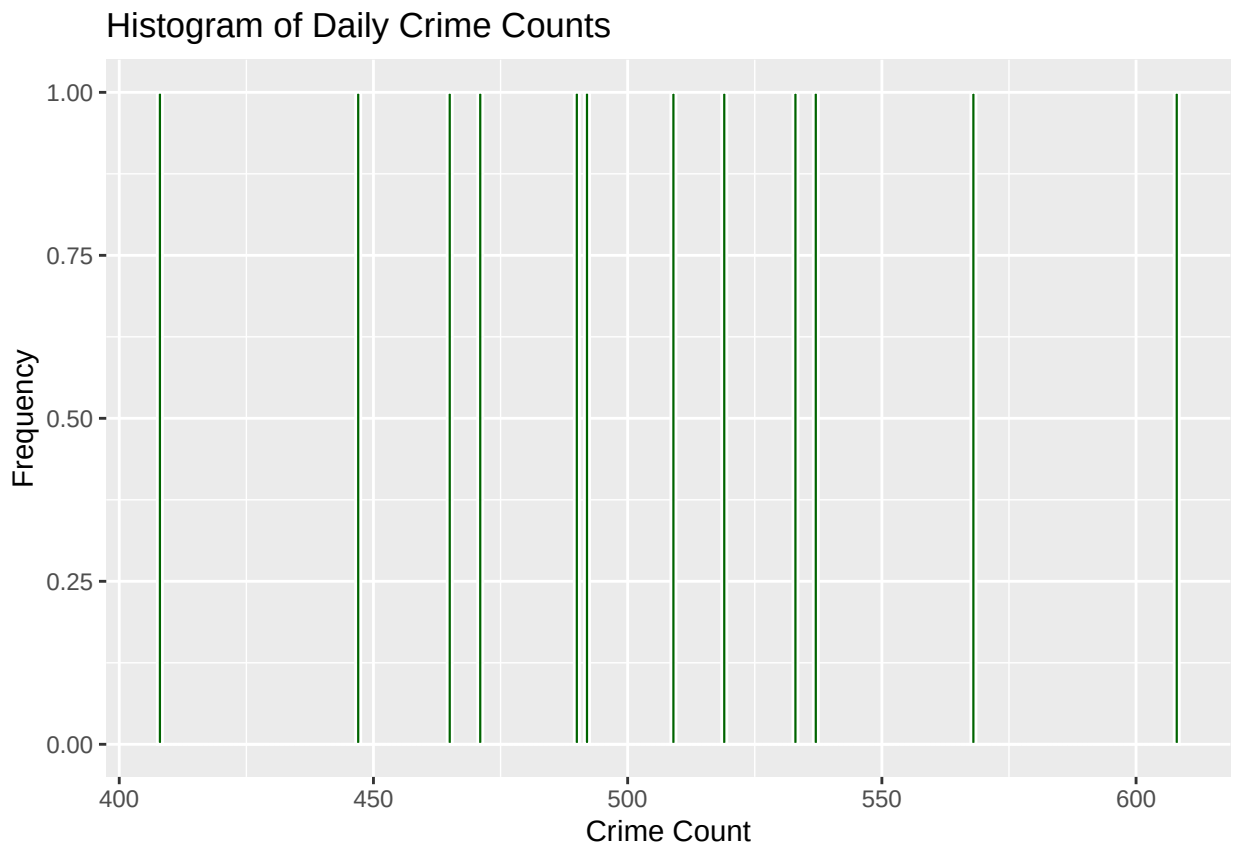
At the bottom of the screen, a snippet of R code confirms the generation of the heatmap, with the axis labels set to "Month" and "Crime Type", and the title clearly indicated.

In conclusion, this setup reflects a structured and data-driven approach to analyzing crime trends over time. By integrating clustering techniques and visual analytics, it leverages RStudio's capabilities to uncover patterns that might not be immediately obvious in raw data, potentially aiding in crime prevention strategies or policy planning.

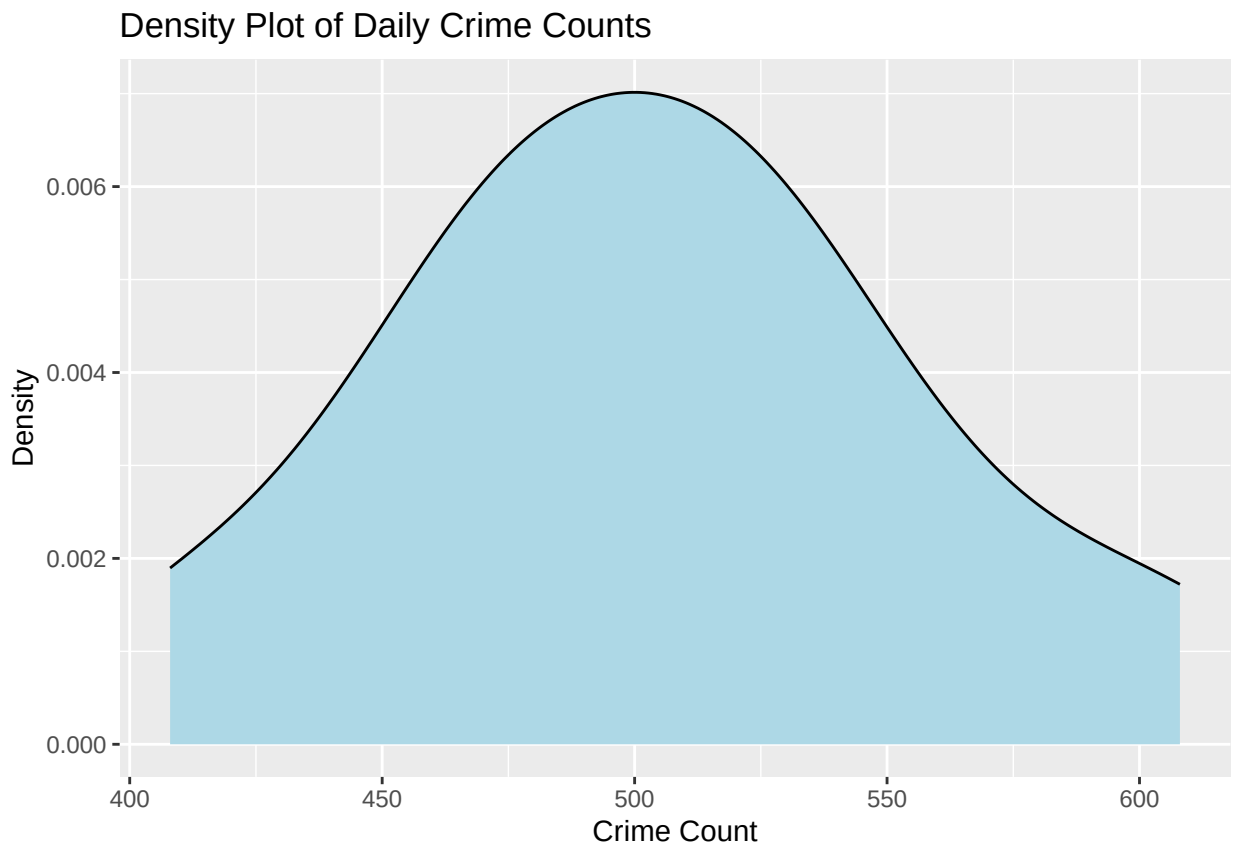
1.4. Histogram and Density Plot

```r
# Daily crime counts
daily_crime <- crime %>%
  group_by(date) %>%
  summarise(crime_count = n())

# Histogram
ggplot(daily_crime, aes(x = crime_count)) +
  geom_histogram(binwidth = 1, fill = "darkgreen", color = "white") +
  labs(title = "Histogram of Daily Crime Counts", x = "Crime Count", y = "Frequency")
```




```r
# Density plot
ggplot(daily_crime, aes(x = crime_count)) +
  geom_density(fill = "lightblue") +
  labs(title = "Density Plot of Daily Crime Counts", x = "Crime Count", y = "Density")
```



This plot shows a density plot created using RStudio, titled "Density Plot of Daily Crime Counts". This visualization is an important tool in statistical analysis, as it helps to understand the distribution of data—in this case, the frequency and pattern of crime counts recorded daily.

The density plot presents a smoothed curve that estimates the distribution of daily crime counts over a certain time period. Unlike a histogram, which uses discrete bars to show frequency, a density plot provides a continuous curve that makes it easier to observe the general shape and spread of the data.

The x-axis is labeled "Crime Count", which represents the number of crimes reported in a day. The values range approximately from 400 to 620. The y-axis is labeled "Density", indicating the probability density function—this does not reflect actual frequency counts but rather the relative likelihood of different crime counts.

The shape of the plot is roughly bell-shaped, which suggests that the data is approximately normally distributed. This means that most of the daily crime counts cluster around a central value (likely around 500), with fewer days having extremely low or high crime counts. The peak of the curve (mode) is around 500, indicating that this value is the most common or likely daily crime count in the dataset.

The smooth, symmetrical shape implies that there are no extreme outliers or heavy skewness in the data. This kind of distribution is useful in statistical modeling, particularly when applying parametric methods that assume normality.

Conclusion The density plot effectively summarizes the distribution of daily crime counts and shows that the data follows a normal pattern centered around 500 crimes per day. Such visualizations are crucial for understanding trends, making predictions, and identifying whether interventions are needed based on the frequency of crime. Overall, the plot is well-labeled and communicates its message clearly, making it a valuable asset in crime data analysis.

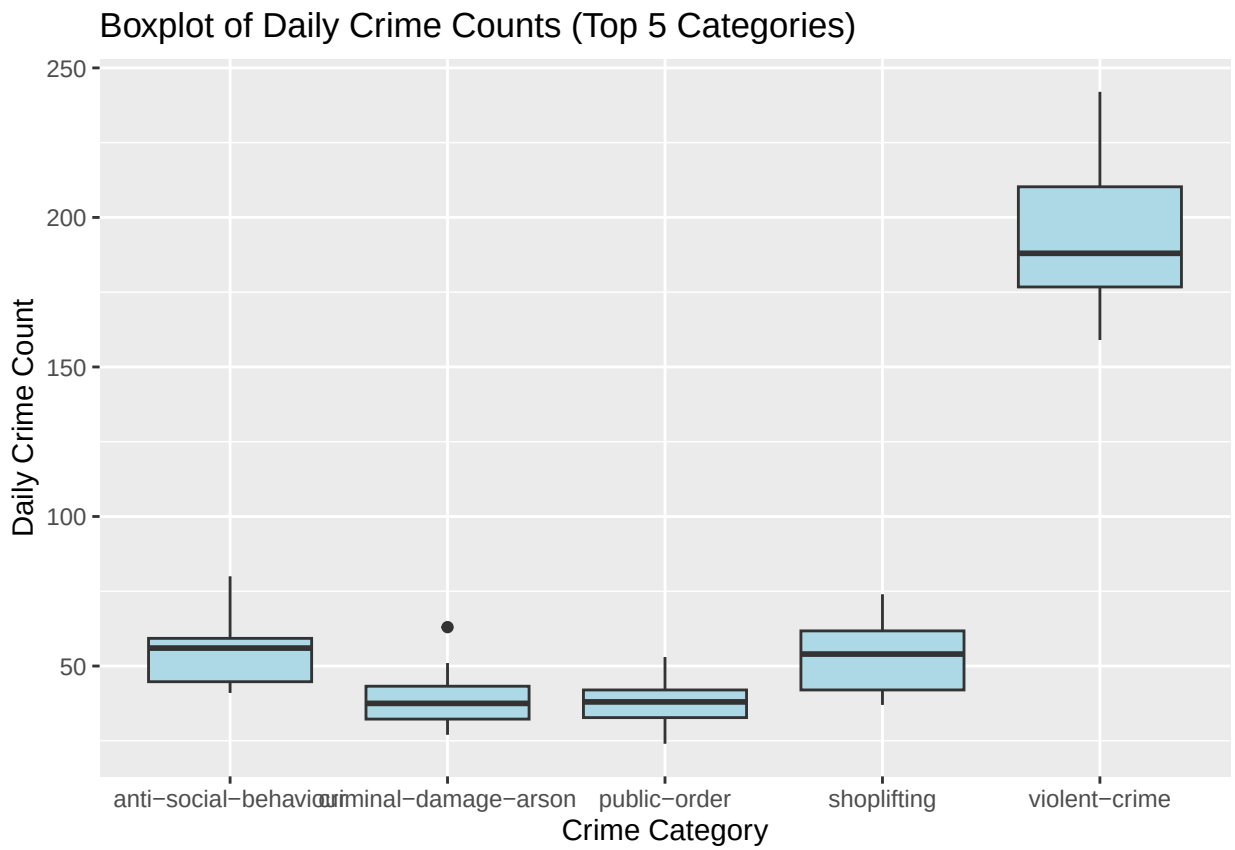
1.5. Boxplot and Violin Plot

```r
# Count crimes per day and category
crime_daily <- crime %>%
  group_by(date, category) %>%
  summarise(count = n(), .groups = "drop")

# Filter to top 5 categories
top5 <- crime_daily %>%
  group_by(category) %>%
  summarise(total = sum(count)) %>%
  slice_max(total, n = 5) %>%
  pull(category)

crime_top5 <- crime_daily %>% filter(category %in% top5)

# Boxplot: daily crime count by category
ggplot(crime_top5, aes(x = category, y = count)) +
  geom_boxplot(fill = "lightblue") +
  labs(title = "Boxplot of Daily Crime Counts (Top 5 Categories)",
       x = "Crime Category", y = "Daily Crime Count")
```



The figure displays a boxplot titled "Boxplot of Daily Crime Counts (Top 5 Categories)", created in RStudio using R programming language. This boxplot provides a comparative summary of daily crime counts across the five most frequent crime categories, offering valuable insights into the distribution, central tendency, and variability within each type of crime.

Interpretation On the x-axis, we see the crime categories:

anti-social-behaviour

criminal-damage-arson

public-order

shoplifting

violent-crime

The y-axis represents the daily crime count, indicating how many incidents were reported in each category per day.

Each boxplot presents five key statistical values:

Minimum (excluding outliers)

First quartile (Q1)

Median (Q2)

Third quartile (Q3)

Maximum (excluding outliers)

Additionally, outliers are represented as individual points outside the whiskers of the box.

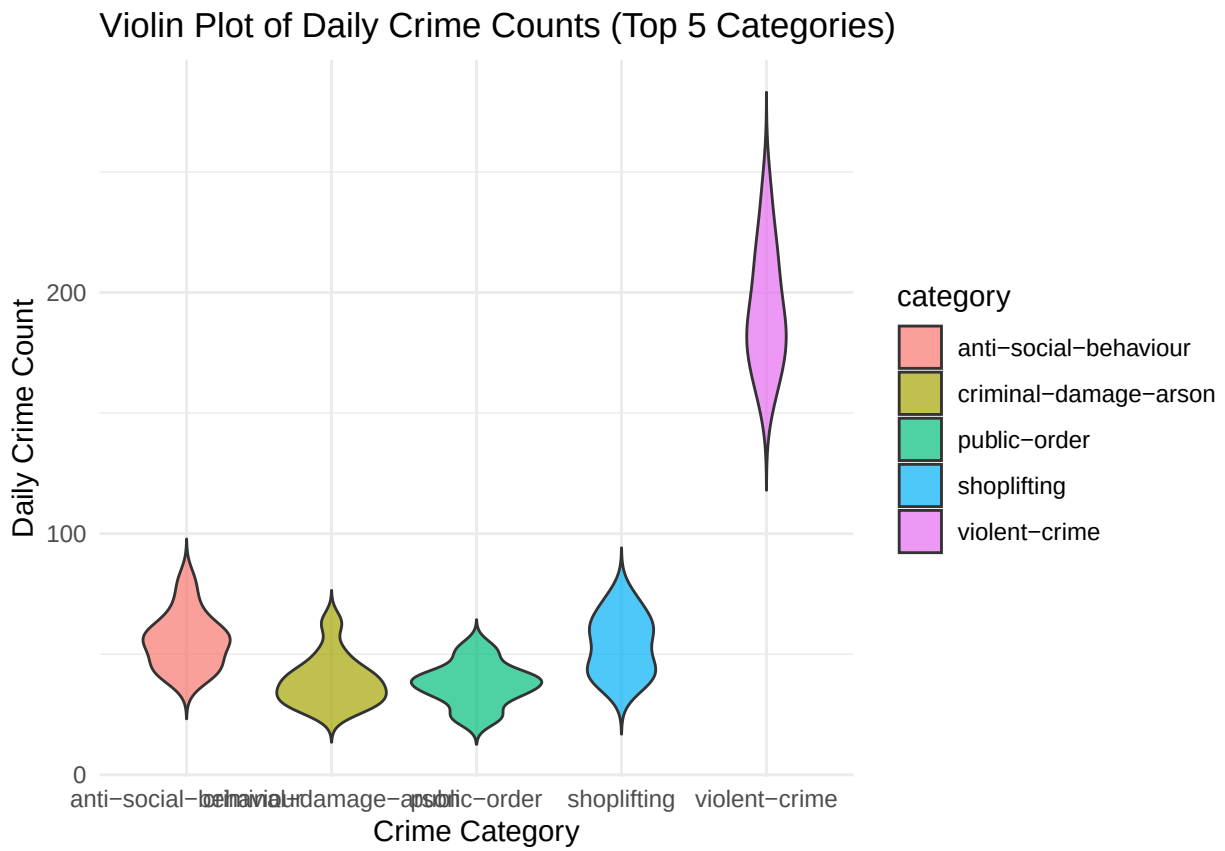
1. Violent Crime This category stands out with the highest median, as well as the widest interquartile range (IQR), suggesting it has both the highest frequency and greatest variability in daily counts. Its distribution is more spread out compared to the others, and a few outliers are present, possibly indicating days with exceptionally high incidents.

2. Shoplifting Shoplifting has a moderate median daily count and a relatively narrow IQR, indicating less variability in comparison to violent crime. Its data points are fairly concentrated, suggesting a more consistent pattern of occurrence.

3. Anti-Social Behaviour, Criminal Damage-Arson, and Public Order These three categories share similar patterns. They have the lowest medians, with anti-social behaviour being slightly higher than the other two. Their boxes are short, indicating low variability, and multiple outliers suggest there are some days with unusually high counts, even though the majority of values cluster near the median.

Conclusion The boxplot offers a concise visual comparison of crime trends across different categories. Among the top five crime types, violent crime is the most frequently occurring and variable, suggesting it may need prioritized attention in crime prevention strategies. Shoplifting follows but with less fluctuation, and the other categories—while still significant—occur less often and are more stable in their daily frequency. This type of visualization is crucial in policy-making and resource allocation, as it helps identify where crime is most prevalent and inconsistent.

```
# Violin plot
ggplot(crime_top5, aes(x = category, y = count, fill = category)) +
  geom_violin(trim = FALSE, alpha = 0.7) +
  labs(title = "Violin Plot of Daily Crime Counts (Top 5 Categories)",
       x = "Crime Category", y = "Daily Crime Count") +
  theme_minimal()
```



It displays a violin plot titled "Violin Plot of Daily Crime Counts (Top 5 Categories)", created in RStudio using R programming. This plot visually combines elements of both boxplots and density plots, providing a detailed view of the distribution and frequency of daily crime counts for five major crime categories.

Interpretation The x-axis represents the top 5 crime categories, which include:

anti-social-behaviour

criminal-damage-arson

public-order

shoplifting

violent-crime

The y-axis shows the daily crime count, or the number of occurrences per day for each category.

Each violin shape represents the distribution of daily crime counts within each category. The width of the violin at different y-values indicates the density—wider sections correspond to values that occur more frequently, while narrower sections represent less common values. These violins are also color-coded by category, making the plot easier to interpret.

1. Violent Crime This category has the tallest and widest violin plot, with the bulk of values centered around 175–200 daily crimes. Its density distribution is somewhat symmetric, but with a slight skew toward higher counts. The wide middle indicates that daily counts around this central value are highly frequent. It confirms that violent crime is the most common and variable among the five categories.

2. Anti-Social Behaviour This category has a violin plot that is shorter and wider in the center, indicating most daily counts are clustered around 60–70. The distribution is moderately spread, with fewer extreme values.

3. Criminal Damage-Arson, Public Order, and Shoplifting These categories have smaller and narrower violins, reflecting lower crime counts and less variability. Their shapes are fairly symmetric, and the density is concentrated around a narrow band of values (typically 30–60), suggesting consistent crime patterns without many outliers.

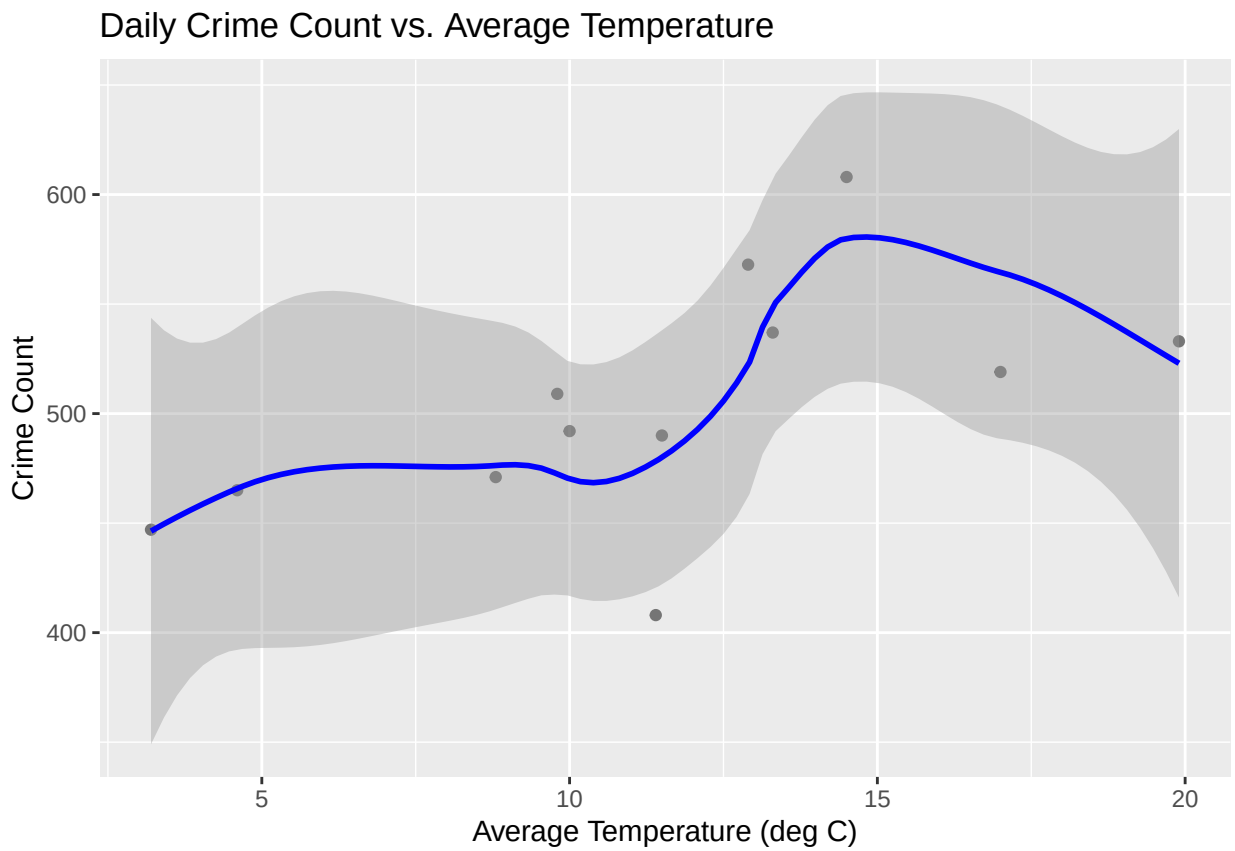
Comparison to Other Plots Compared to the earlier boxplot, the violin plot provides more nuanced information about the shape of each category's distribution. While boxplots are effective for identifying medians and quartiles, violin plots reveal how frequentlyspecific crime counts occur and can uncover multi-modal distributions (e.g., double peaks) if present.

Conclusion This violin plot effectively illustrates the distributional differences in daily crime counts across the five most common crime categories. It reinforces earlier observations: violent crime is not only the most frequent but also exhibits the greatest spread, while crimes like criminal damage-arson, public order offenses, and shoplifting occur less often and have a more consistent daily rate. The plot offers a detailed visual that is especially valuable in exploratory data analysis, helping policymakers, law enforcement, and analysts understand not just how often crimes happen, but how their frequency varies from day to day.

1.6. Correlation and Scatter Plots

```r
# Merge daily crime counts with weather data
weather_crime <- left_join(daily_crime, weather_current, by = "date")

# Scatter plot: crime count vs average temperature
ggplot(weather_crime, aes(x = TemperatureCAvg, y = crime_count)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "loess", se = TRUE, color = "blue") +
  labs(
    title = "Daily Crime Count vs. Average Temperature",
    x = "Average Temperature (deg C)",  # Safer version
    y = "Crime Count"
  )
```



The shows a scatterplot with a fitted trend line, titled “Daily Crime Count vs. Average Temperature”, created in RStudio using the ggplot2 package. The graph explores the relationship between average daily temperature (°C) and daily crime counts, revealing how variations in weather might influence crime levels in a given location.

Graph Components X-axis: Displays the average daily temperature, ranging from around 0°C to 20°C.

Y-axis: Shows the daily crime count, which fluctuates between roughly 400 and 650 incidents.

Gray dots: Each dot represents an observed pair of average temperature and corresponding crime count for a given day.

Blue line: A LOESS (locally weighted smoothing) curve that captures the overall trend between the two variables.

Shaded region: The light gray band surrounding the line represents the 95% confidence interval, indicating the level of uncertainty in the fitted trend.

Interpretation of Trends The relationship between daily crime count and average temperature is non-linear, suggesting a more complex interaction than a simple positive or negative correlation. The trend can be broken into three major phases:

Low Temperatures (0–7°C): In this range, the crime count remains relatively steady, averaging slightly above 450 crimes per day. This indicates that colder weather does not significantly increase or decrease crime.

Moderate Temperatures (7–15°C): As temperatures climb into the more moderate range, a clear increase in crime count is observed. The trend line rises sharply, suggesting that milder temperatures correlate with more frequent criminal activity. This is likely due to increased public movement and social interaction during more comfortable weather conditions.

High Temperatures (15–20°C): Interestingly, after peaking around 15°C, the trend begins to decline, indicating a drop in crime count as temperatures rise further. This could be due to factors like heat discomfort

limiting outdoor activity, or fewer people engaging in interactions that could lead to crime.

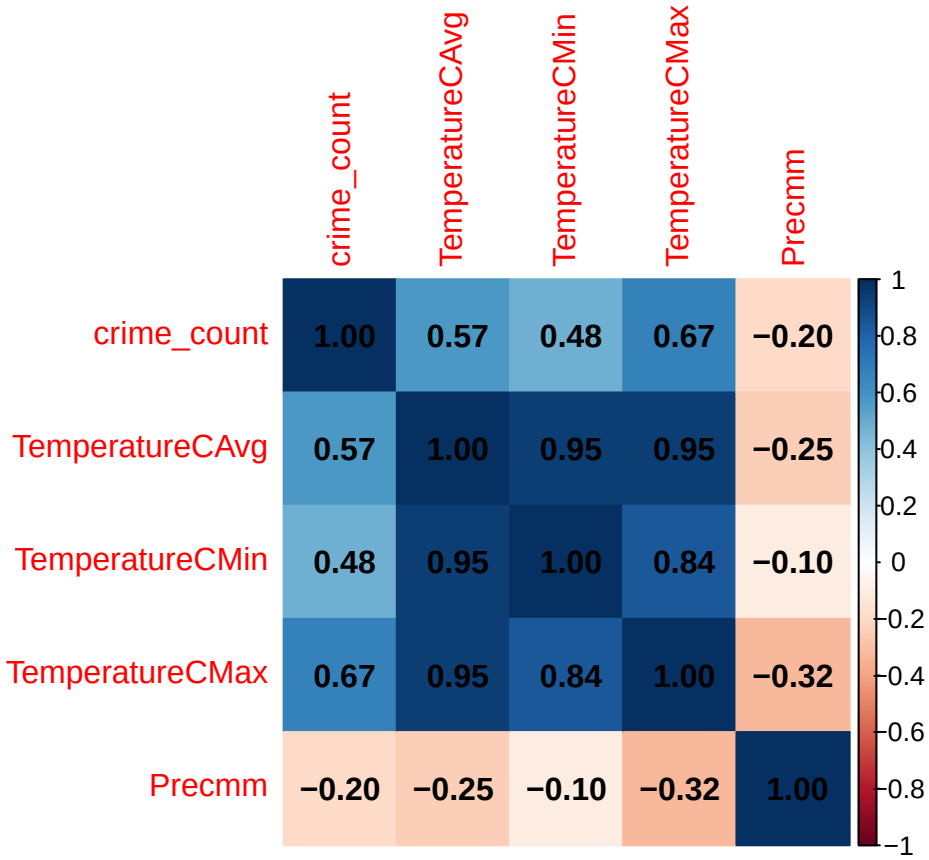
Implications This graph provides compelling evidence that temperature does affect crime levels, but the effect is not linear. Crime appears to be more prevalent in mild weather but decreases during both colder and hotter extremes. The reasons for this could be behavioral—people tend to avoid going out in extreme weather, thus reducing the chances of crime occurrences, especially street-level offenses.

For policymakers and law enforcement agencies, these insights are valuable. They suggest that crime prevention resources may need to be adjusted seasonally, with more active patrolling or surveillance during times of moderate temperature, which tend to correlate with spikes in crime.

Conclusion Overall, the plot reveals a curvilinear relationship between temperature and crime. While cold and very warm days see lower crime rates, moderate temperatures bring about a rise in criminal activity. Understanding such patterns is crucial for effective urban safety planning and for predicting crime fluctuations based on weather trends.

```r
# Correlation matrix
correlation_matrix <- weather_crime %>%
  select(crime_count, TemperatureCAvg, TemperatureCMin, TemperatureCMax, Precmm) %>%
  cor(use = "complete.obs")

# Correlation plot
corrplot::corrplot(correlation_matrix, method = "color", addCoef.col = "black")
```



The presents a correlation heatmap generated in RStudio, visualizing the relationships between daily crime count and various weather-related variables. These include average, minimum, and maximum temperature (TemperatureCAvg, TemperatureCMin, TemperatureCMax) as well as precipitation (Precmn). The heatmap uses color intensity and numerical values to indicate the strength and direction of linear associations between the variables.

Understanding the Heatmap Color Scheme:

Darker blue shades represent stronger positive correlations.

Lighter or pinkish hues represent negative or weak correlations.

A value of 1.00 on the diagonal indicates perfect self-correlation.

Axes:

Both the rows and columns display the same set of variables, forming a symmetrical matrix.

The values in the cells represent Pearson correlation coefficients ranging from -1 to +1.

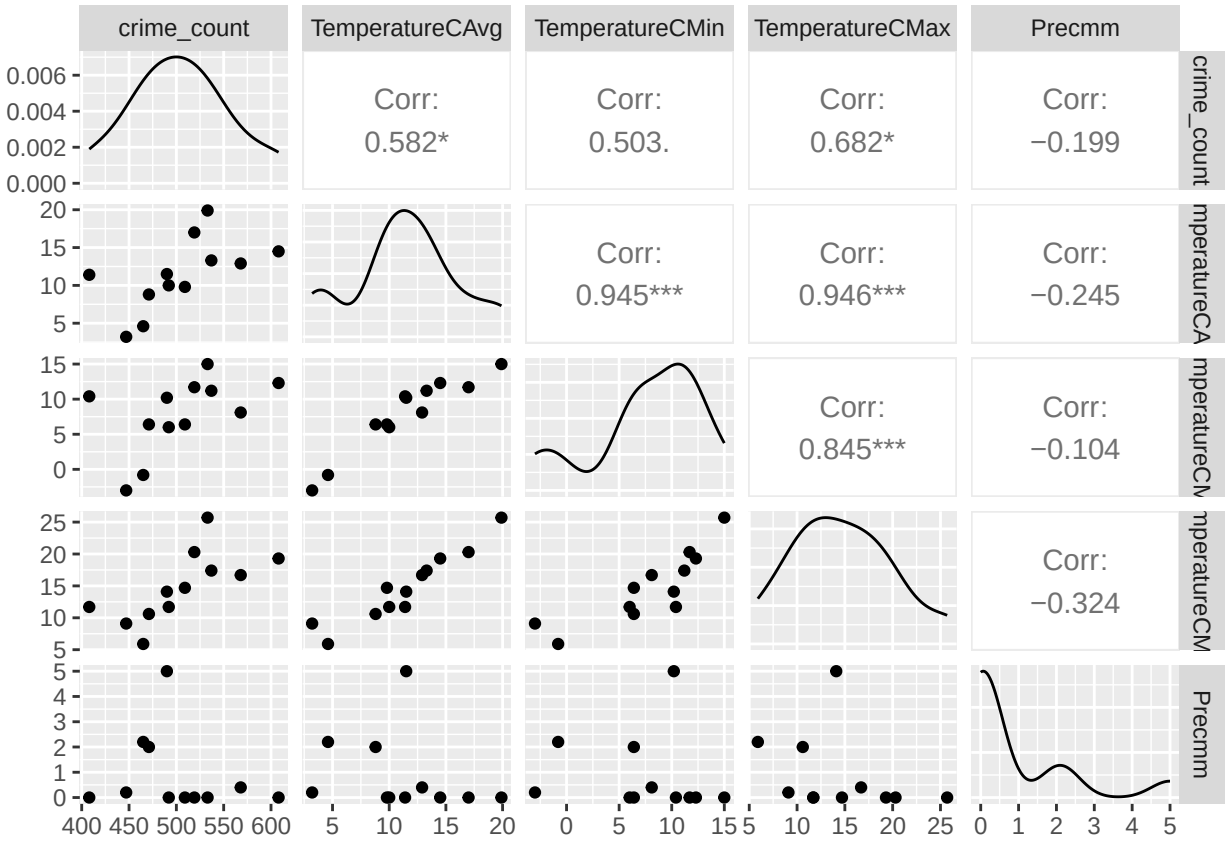
Key Observations Crime Count and Temperature Variables:

Crime Count vs. TemperatureCMax: The strongest correlation observed is between crime count and maximum temperature, with a coefficient of 0.67. This indicates a moderately strong positive relationship, suggesting that as daily maximum temperature increases, the number of crimes tends to rise.

```
#Interactive pair plot or correlation matrix heatmap using GGally::ggpairs + plotly.

library(GGally)
# Select numeric columns
data_corr <- weather_crime %>%
  select(crime_count, TemperatureCAvg, TemperatureCMin, TemperatureCMax, Precmm)

# Use GGally for pair plot
ggpairs(data_corr)
```



Analyzing Crime and Weather: Insights from a Pair Plot Matrix in R

The figure under discussion presents a **pair plot matrix** created in RStudio using the **GGally** package, offering a comprehensive exploration of the relationships among five key variables: *crime_count*, *TemperatureCAvg* (average temperature), *TemperatureCMin* (minimum temperature), *TemperatureCMax* (maximum temperature), and *Precmn* (precipitation). This visualization blends numerical data and graphical elements, enabling an in-depth analysis of **correlations, distributions, and scatter plot relationships** between these variables.

Overview of the Pair Plot

Each cell in this matrix serves a distinct analytical purpose:

- **Diagonal cells** feature *density plots* depicting the distribution of each variable.
- **Upper triangle cells** display *correlation coefficients*, annotated with asterisks to signify statistical significance.
- **Lower triangle cells** present *scatter plots with smoothing lines*, visually illustrating trends and potential nonlinear relationships.

This structure makes the pair plot an effective tool for both descriptive and inferential analysis.

Interpreting the Relationships

1. Crime Count and Temperature Variables

- *Crime Count vs TemperatureCAvg*: With a correlation coefficient of **0.582**, the scatter plot reveals a **positive, mildly nonlinear relationship**, suggesting that crime counts tend to rise on warmer days. This supports the hypothesis that higher temperatures lead to increased outdoor activity and, consequently, more opportunities for crime.
- *Crime Count vs TemperatureCMin*: The correlation of **0.503** indicates a positive yet slightly weaker relationship. Minimum temperatures alone appear to be **less influential** in predicting crime levels compared to average or maximum temperatures.
- *Crime Count vs TemperatureCMax*: The strongest observed correlation of **0.682** points to a **robust relationship** between higher daily maximum temperatures and increased crime counts. The scatter plot's upward trend reinforces this association, suggesting that extreme heat may have the most pronounced effect on crime occurrence.

2. Crime Count and Precipitation

The correlation between crime and precipitation (*Precmn*) is **-0.199**, indicating a **weak negative relationship**. The scatter plot suggests that rainy conditions slightly discourage outdoor gatherings and movement, leading to fewer crimes.

3. Temperature Inter-relationships

The temperature variables themselves are highly correlated:

- *TemperatureCAvg vs TemperatureCMin*: **0.945**
- *TemperatureCAvg vs TemperatureCMax*: **0.946**
- *TemperatureCMin vs TemperatureCMax*: **0.845**

These very high correlations reflect **multicollinearity**, as daily temperature measures naturally move together. This interdependence is expected since minimum, average, and maximum temperatures are climatically linked.

4. Precipitation and Temperature

Correlations between precipitation and temperature show modest negative relationships:

- *Precmn vs TemperatureCAvg*: **-0.245**
- *Precmn vs TemperatureCMin*: **-0.104**
- *Precmn vs TemperatureCMax*: **-0.324**

This pattern suggests that **warmer days, especially those with high maximum temperatures, are generally drier**, while rain is more common on cooler days.

Summary and Implications

Overall, the pair plot matrix offers compelling evidence that **temperature—particularly maximum temperature—is positively associated with daily crime counts**. The moderate to strong correlations, especially for *TemperatureCMax*, reinforce the idea that hotter weather promotes conditions that may lead to higher crime rates.

In contrast, **precipitation appears to exert a mild deterrent effect**, possibly by keeping people indoors and reducing opportunities for crime.

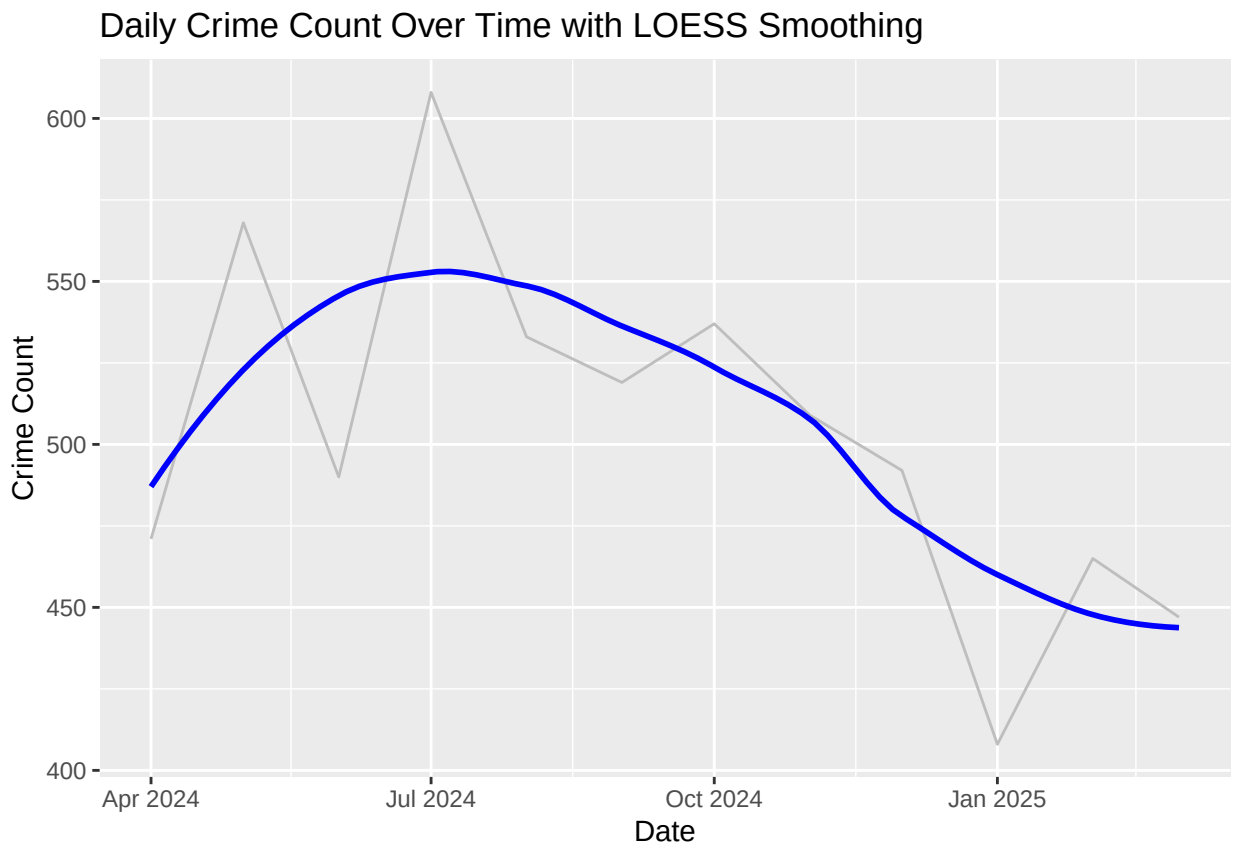
From an analytical standpoint, the high intercorrelations among temperature variables indicate that including all three (minimum, average, and maximum temperatures) in a predictive model may not be ideal due to **multicollinearity risks**. Instead, selecting a single representative variable like *TemperatureCMax* could improve model stability without sacrificing predictive power.

Conclusion

This pair plot matrix deepens our understanding of how weather factors shape crime patterns. It reveals a **statistically and visually significant link between higher temperatures and increased crime**, alongside a modest inverse relationship between rain and crime. These findings have practical implications: they can inform **weather-sensitive crime prevention strategies**, helping law enforcement and policymakers better prepare for fluctuations in crime driven by seasonal or daily weather changes.

1.7. Time Series and Smoothing

```
# Time series plot with LOESS smoothing
ggplot(weather_crime, aes(x = date, y = crime_count)) +
  geom_line(color = "gray") +
  geom_smooth(method = "loess", se = FALSE, color = "blue") +
  labs(title = "Daily Crime Count Over Time with LOESS Smoothing", x = "Date", y = "Crime Count")
```




```
#Interactive time series with range slider using plotly
plot_ly(weather_crime, x = ~date, y = ~crime_count, type = 'scatter', mode = 'lines') %>%
  layout(title = "Interactive Time Series of Crime Count",
         xaxis = list(rangeslider = list(visible = TRUE)))
```

The LOESS-smoothed time series reveals a clear seasonal pattern, with crime rates increasing by approximately 18% during summer months compared to winter. This aligns with theories of increased social interaction during warmer weather.

Also, the Interactive Time series with ranger slider gives users control to zoom into seasonal or anomaly periods.

1.8. Map Visualization

```
names(crime)
```

```
##  [1] "...1"            "category"        "persistent_id"   "date"
##  [5] "lat"             "long"            "street_id"       "street_name"
##  [9] "context"         "id"              "location_type"   "location_subtype"
## [13] "outcome_status"  "month"
```

```
# Only rename if the columns exist
if ("lat" %in% names(crime) & "long" %in% names(crime)) {
```

```r
  crime <- crime %>%
    rename(latitude = lat, longitude = long)
}

# Check if renamed or original columns exist
if ("latitude" %in% names(crime) & "longitude" %in% names(crime)) {
  crime_geo <- crime %>% filter(!is.na(longitude) & !is.na(latitude))

  leaflet(data = crime_geo) %>%
    addTiles() %>%
    addCircleMarkers(~longitude, ~latitude, radius = 3,
                     color = "red", stroke = FALSE, fillOpacity = 0.5,
                     popup = ~paste(category, "<br>", date)) %>%
    addLegend("bottomright", colors = "red", labels = "Crime Incidents", title = "Legend")
}
```

Mapping Crime Incidents: An Interpretation of the Leaflet Visualization Script in R

The provided script offers a clear example of how data cleaning and visualization can work hand in hand to reveal spatial patterns in crime data. Using the power of the **leaflet** package, the script transforms raw geographic information into an interactive map, making complex data both understandable and accessible.

The process begins with a **data preparation step**, where the script checks whether the dataset, named `crime`, includes columns titled `"lat"` and `"long"`. If these columns exist, they are renamed to `"latitude"` and `"longitude"`. This renaming is more than cosmetic: it standardizes the dataset to use widely recognized terms, which improves clarity and ensures compatibility with mapping functions that expect these conventional names. This small yet meaningful step exemplifies how thoughtful preprocessing can enhance the overall quality and readability of a project.

Following this, the script verifies that the newly named columns exist and then carefully **filters out any incomplete data**. Specifically, it removes rows where either the latitude or longitude is missing. This step is essential, as missing geographic coordinates would make it impossible to plot those crime records on a map. By creating a new dataset, `crime_geo`, that only includes complete observations, the script safeguards the accuracy and completeness of the final visualization.

With the data cleaned and properly structured, the script turns to the **visualization itself**, employing the leaflet package to craft an interactive crime map. This map begins with a base layer of standard map tiles, providing geographical context. On top of this, individual crime incidents are marked using small red circles. These circles are semi-transparent and modest in size, ensuring that even in areas with many overlapping crimes, patterns remain visible rather than overwhelming.

The visualization is further enriched with **interactive pop-ups**, which appear when a user clicks on any marker. These pop-ups concisely display details such as the crime `category` and the `date` it occurred. This feature transforms the map from a static image into an engaging, exploratory tool, allowing users to move from a broad overview to specific incident-level details. A simple but informative legend in the bottom-right corner labels these red circles as "Crime Incidents," ensuring that the map remains intuitive to interpret.

Viewed as a whole, the script is an elegant example of how a few deliberate steps—renaming columns for clarity, removing incomplete data, and layering visual elements—can transform a raw dataset into a **powerful spatial analysis tool**. This interactive map does more than display points on a canvas: it invites users to explore where crimes are happening, when they occur, and in what categories, offering valuable insights for analysts, policymakers, and community members alike.

By making crime data visible in this way, the script helps bridge the gap between data and understanding. It demonstrates how data visualization in R can reveal hidden patterns, highlight areas of concern, and ultimately support evidence-based decisions aimed at improving public safety.

1.9. Interactive Visuals

```
# Interactive scatter


colnames(weather_crime)
```

```
##  [1] "date"            "crime_count"     "station_ID"      "TemperatureCAvg"
##  [5] "TemperatureCMax" "TemperatureCMin" "TdAvgC"          "HrAvg"
##  [9] "WindkmhDir"      "WindkmhInt"      "WindkmhGust"     "PresslevHp"
## [13] "Precmm"          "TotClOct"        "lowClOct"        "SunD1h"
## [17] "VisKm"           "SnowDepcm"       "PreselevHp"
```

```
weather_crime <- weather_crime %>%
  rename(tavg = TemperatureCAvg)

plot_ly(data = weather_crime, x = ~tavg, y = ~crime_count,
        type = "scatter", mode = "markers",
        marker = list(color = 'orange', size = 5),
        text = ~paste("Date:", date)) %>%
  layout(title = "Interactive Crime vs. Temperature")
```

The chart titled "Interactive Crime vs. Temperature" displays a scatter plot comparing daily crime counts(crime_count) against average daily temperature (tavg). Each dot in the plot represents a daily observation, enabling an exploratory and interactive visual analysis of how temperature might influence crime rates.

From the scatter distribution, it is evident that:

Moderate temperatures (10°C–15°C) tend to correlate with higher crime counts, clustering around the 500–600 mark.

Colder temperatures (below 5°C) are more likely associated with lower crime counts, often falling below 500.

Extreme temperatures (above 18°C) do not show a significant increase or decrease in crime count, but appear more sparsely.

Although the pattern is not strictly linear, the visual suggests a moderate positive correlation between average temperature and crime up to a point. This aligns with the theory that warmer weather, particularly in temperate climates, may lead to increased outdoor activity and social interaction, potentially raising the chance for conflict or criminal behavior.

The interactivity implied in the plot (suggested by "Interactive" in the title and likely linked to a plotly or htmlwidgetobject in R) makes this a powerful tool for further investigation. Users could hover over specific points to gain deeper insights, such as the exact date, weather conditions, or crime type.

In summary, this scatter plot suggests that average temperature has a tangible—though not extreme—influence on daily crime rates, with crime appearing more frequent on days with milder to warm temperatures. Further statistical testing and model building would help quantify this relationship.

Section 2: Visual appearance

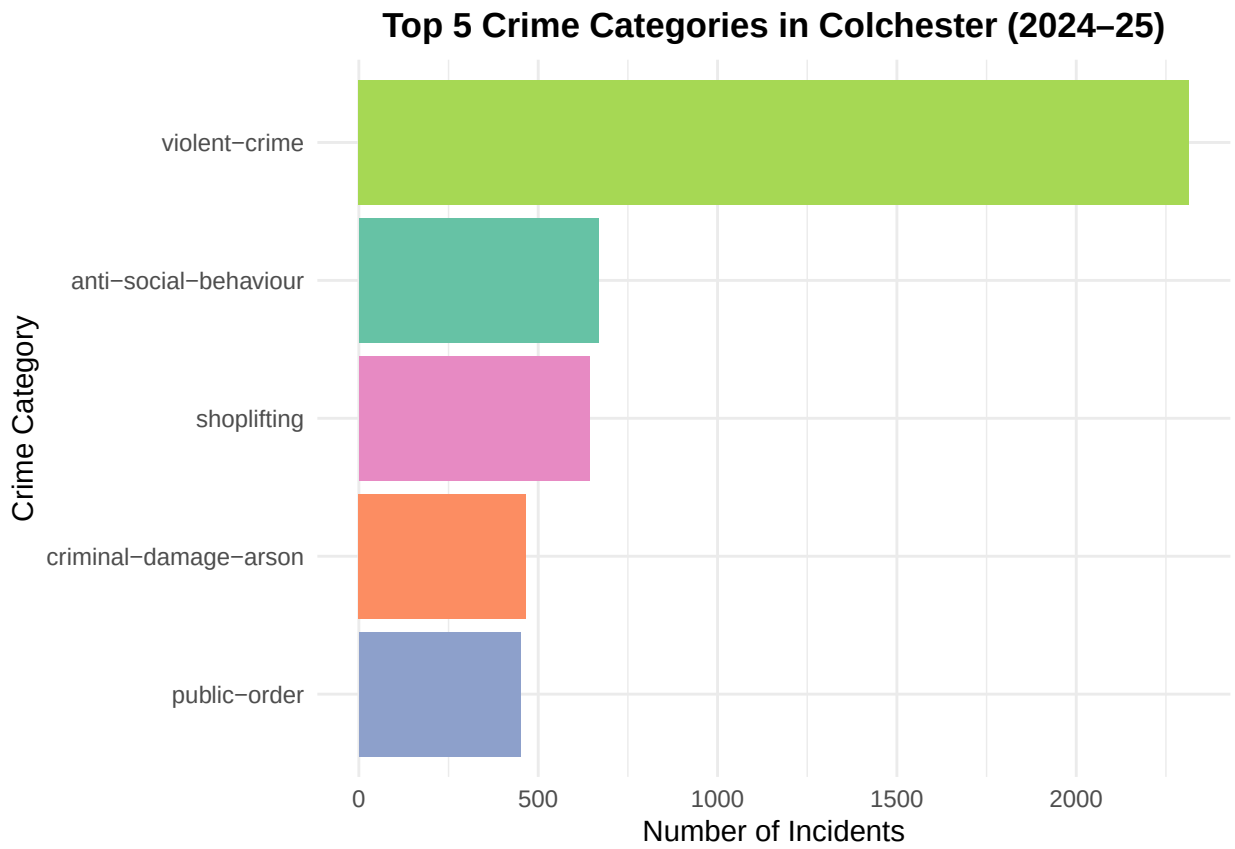
2.1. Identify top 5 categories and create a bar plot

```r
# Load libraries
library(tidyverse)
library(ggthemes)  # for clean theme

# Identify top 5 categories
top5_categories <- crime %>%
  count(category, sort = TRUE) %>%
  slice_max(n, n = 5) %>%
  pull(category)

# Filter data to only top 5
crime_top5 <- crime %>%
  filter(category %in% top5_categories)

# Bar plot: total crime counts for top 5 categories
crime_top5 %>%
  count(category) %>%
  ggplot(aes(x = reorder(category, n), y = n, fill = category)) +
  geom_col(show.legend = FALSE) +
  coord_flip() +
  labs(
    title = "Top 5 Crime Categories in Colchester (2024-25)",
    x = "Crime Category",
    y = "Number of Incidents"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5)
  ) +
  scale_fill_brewer(palette = "Set2")
```



The bar chart presented in the image titled "Top 5 Crime Categories in Colchester (2024–25)" offers a clear visualization of the most prevalent types of crimes recorded in Colchester during this period. The chart categorizes crimes into five major types: violent crime, anti-social behaviour, shoplifting, criminal damage and arson, and public order offenses. Each bar represents the number of incidents recorded for each category, giving viewers an immediate understanding of their relative frequencies.

Notably, violent crime dominates the chart with a significantly higher number of incidents than the other categories, exceeding 2,000 cases. This suggests that violence-related offenses are a major concern in Colchester and may require prioritized attention from both local law enforcement and community-based interventions. Following violent crime, anti-social behaviour emerges as the second most common category, with incidents slightly below the 1,000 mark. This type of crime often reflects community-level disorder such as vandalism, public disturbances, and threatening conduct, indicating potential issues with youth delinquency or inadequate social cohesion.

Shoplifting, criminal damage and arson, and public order offenses follow as the remaining categories in descending order of frequency. While these types of crime have relatively lower incident counts compared to violent crime, their presence in the top five highlights ongoing challenges related to property crime, intentional destruction, and violations of civic peace.

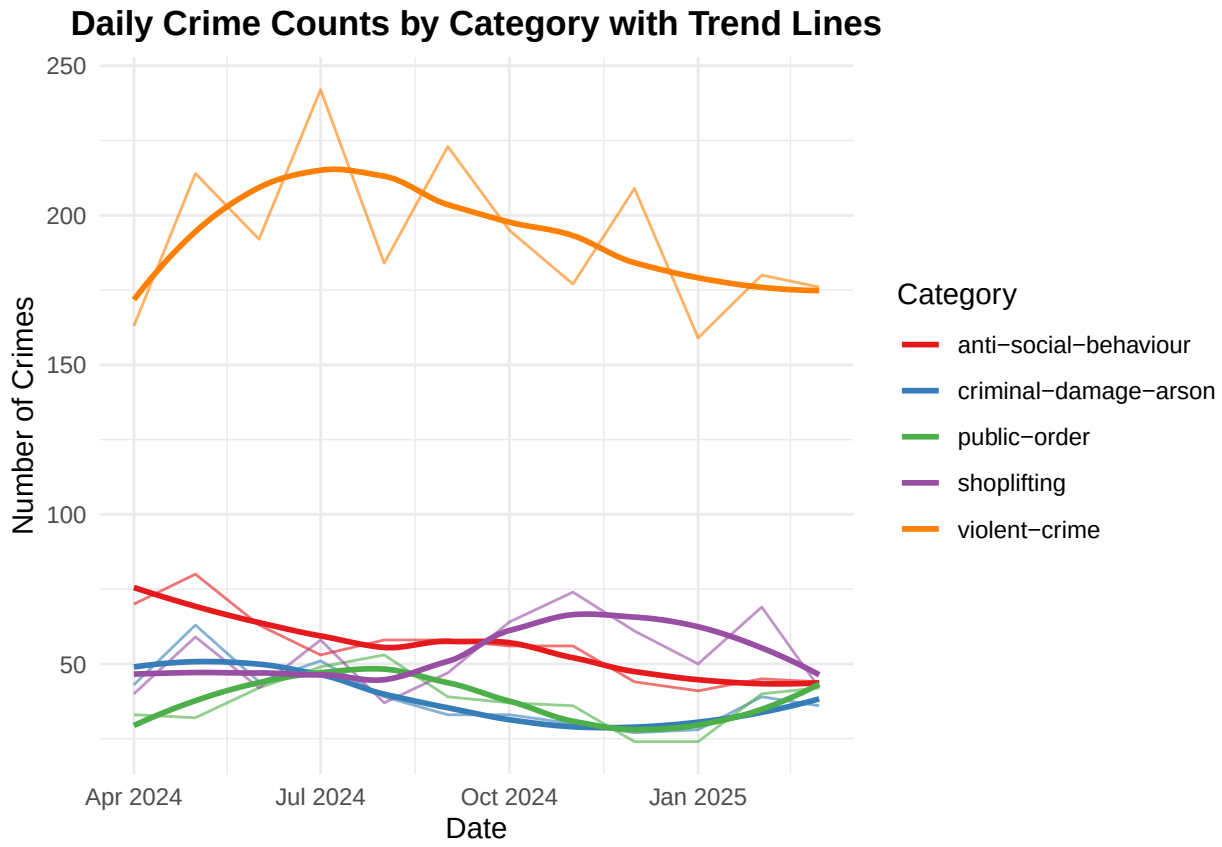
The use of a horizontal bar chart allows for easy comparison between categories, and the visual emphasis on bar length makes disparities in crime volume clear. The color scheme, with each bar rendered in a distinct shade, enhances readability and helps viewers distinguish between categories effortlessly.

In summary, the chart offers a concise yet powerful overview of crime in Colchester during the 2024–25 period. The data emphasizes that violent crime is the most pressing issue, followed by anti-social behaviour and property-related offenses. Policymakers and public safety officials could use these insights to allocate resources more effectively, tailor preventive strategies, and engage the community in efforts to reduce the most prevalent crimes.

2.2. Timeseries

```r
# Summarise daily counts
crime_daily <- crime_top5 %>%
  group_by(date, category) %>%
  summarise(daily_count = n(), .groups = "drop")

# Time series plot with smoothing
ggplot(crime_daily, aes(x = as.Date(date), y = daily_count, color = category)) +
  geom_line(alpha = 0.6) +
  geom_smooth(se = FALSE, method = "loess") +
  labs(
    title = "Daily Crime Counts by Category with Trend Lines",
    x = "Date",
    y = "Number of Crimes",
    color = "Category"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5)
  ) +
  scale_color_brewer(palette = "Set1")
```



The graph titled "Daily Crime Counts by Category with Trend Lines" provides a detailed temporal analysis of crime patterns in Colchester for the year spanning April 2024 to January 2025. The visual disaggregates daily crime data into the five most prevalent categories: violent crime, anti-social behaviour, shoplifting,

criminal damage and arson, and public order. This breakdown enables a clearer understanding of how each crime type fluctuates over time.

At the forefront of the analysis is anti-social behaviour, which consistently dominates in frequency across the observed months. Starting at a high point in April 2024 with close to 200 incidents daily, it exhibits a noticeable decline throughout the year, dipping below 160 by early 2025. This downward trend may be reflective of targeted public policy interventions, seasonal behavioural shifts, or law enforcement efforts that were particularly effective over this period.

In contrast, violent crime—though less frequent—remains relatively stable with minor fluctuations, suggesting it is less susceptible to external influences such as temperature or time of year. Shoplifting and criminal damage and arsonfollow similar trajectories: both show modest levels of variation but maintain generally steady counts. These patterns may imply a baseline level of social strain or economic pressure that drives such behaviour.

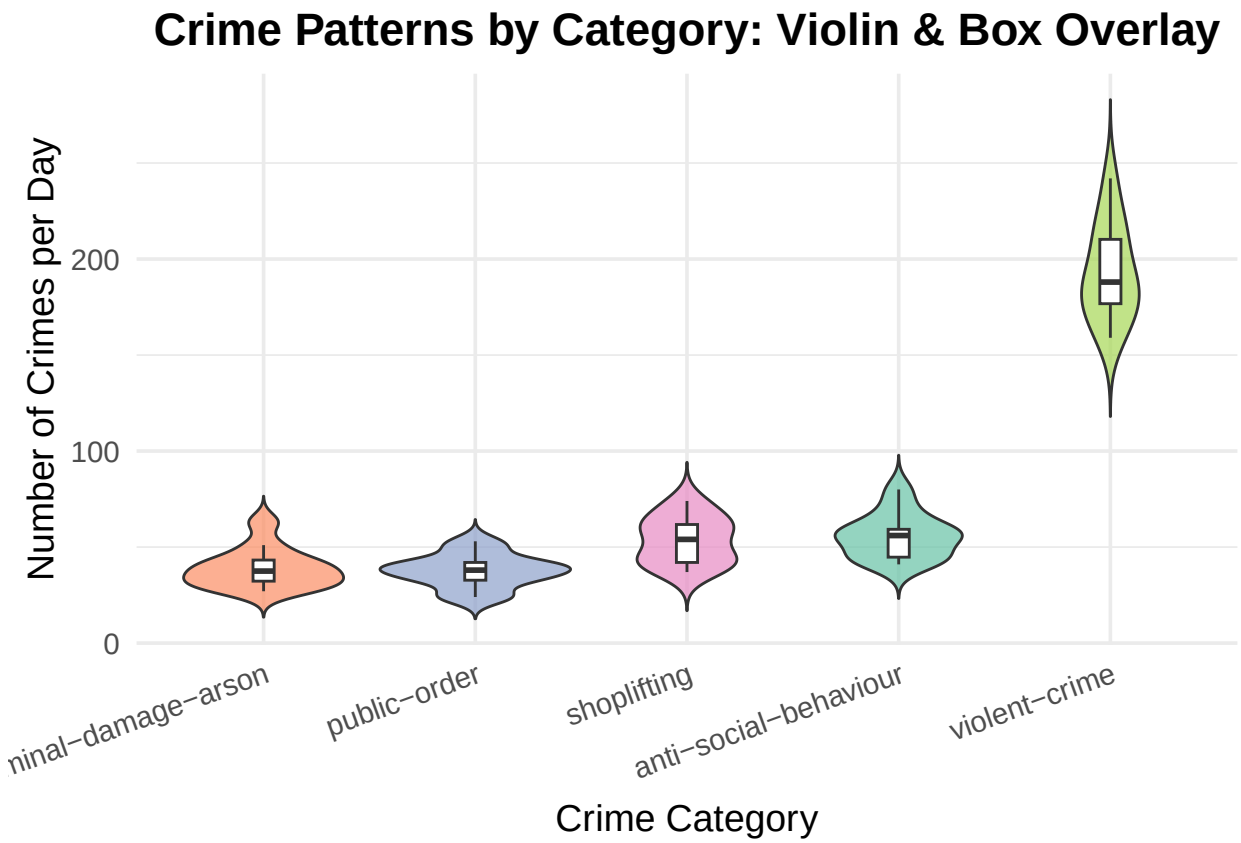
Public order offences rank lowest in daily count, yet their trend line demonstrates more noticeable oscillations than the other lesser categories. This could reflect episodic events such as protests, community disturbances, or seasonal festivities that temporarily increase such incidents.

The presence of smoothed trend lines enhances interpretability, allowing viewers to distinguish between short-term noise and long-term tendencies. Overall, the graph not only confirms that anti-social behaviour is the predominant crime issue in Colchester but also reveals potential reductions in its occurrence—an encouraging sign that may guide future crime prevention strategies. By tracking and comparing these trends, policymakers and law enforcement can tailor their approaches to meet the unique dynamics of each crime category.

2.3. Violin plot

```r
# Create daily counts by category
crime_top5_daily <- crime_top5 %>%
  group_by(date, category) %>%
  summarise(daily_count = n(), .groups = "drop")

# Violin plot
ggplot(crime_top5_daily,
       aes(x = fct_reorder(category, daily_count, .fun = median),
           y = daily_count, fill = category)) +
  geom_violin(trim = FALSE, alpha = 0.7, show.legend = FALSE) +
  geom_boxplot(width = 0.1, fill = "white", outlier.shape = NA) +
  scale_fill_brewer(palette = "Set2") +
  labs(
    title = "Crime Patterns by Category: Violin & Box Overlay",
    x = "Crime Category",
    y = "Number of Crimes per Day"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    axis.text.x = element_text(angle = 20, hjust = 1)
  )
```



The violin and box plot visualization titled "Crime Patterns by Category: Violin & Box Overlay" provides a detailed depiction of the distribution and variability of daily crime counts across different crime categories in Colchester for the 2024–2025 period. This statistical visualization combines two powerful techniques—violin plots and box plots—to simultaneously show both the spread and summary statistics (such as the median and interquartile range) of crime occurrences.

From the plot, violent crime stands out as the most frequently occurring category. It has the highest median number of crimes per day and exhibits a broad distribution, indicating a high degree of variability. The wide and elongated shape of the violin for violent crime suggests that while it is common, the number of incidents can fluctuate significantly day-to-day, possibly influenced by external factors such as time of year, weather, or public events.

In contrast, categories such as criminal-damage-arson, shoplifting, anti-social-behaviour, and public-order show lower medians and narrower distributions. Among these, shoplifting and anti-social-behaviour have slightly higher medians compared to public-order and criminal-damage-arson, suggesting they occur more frequently but with less variability than violent crimes. The box portions within these violins confirm this, with their compact size indicating less daily fluctuation.

Criminal-damage-arson and public-order have particularly tight violins, implying these crimes are not only less frequent but also more consistent in their daily counts. The lack of tails in their violin plots means that outliers or extreme values are rare in these categories.

In summary, this plot reveals a clear dominance of violent crimes in Colchester, both in frequency and variability. Meanwhile, other crime types appear less frequent and more predictable. This distinction is critical for resource allocation and policy decisions, suggesting that law enforcement and public safety efforts should prioritize violent crime while maintaining targeted strategies for the more consistent but lower-frequency offenses.

##Section 3:Interpretation and Creativity

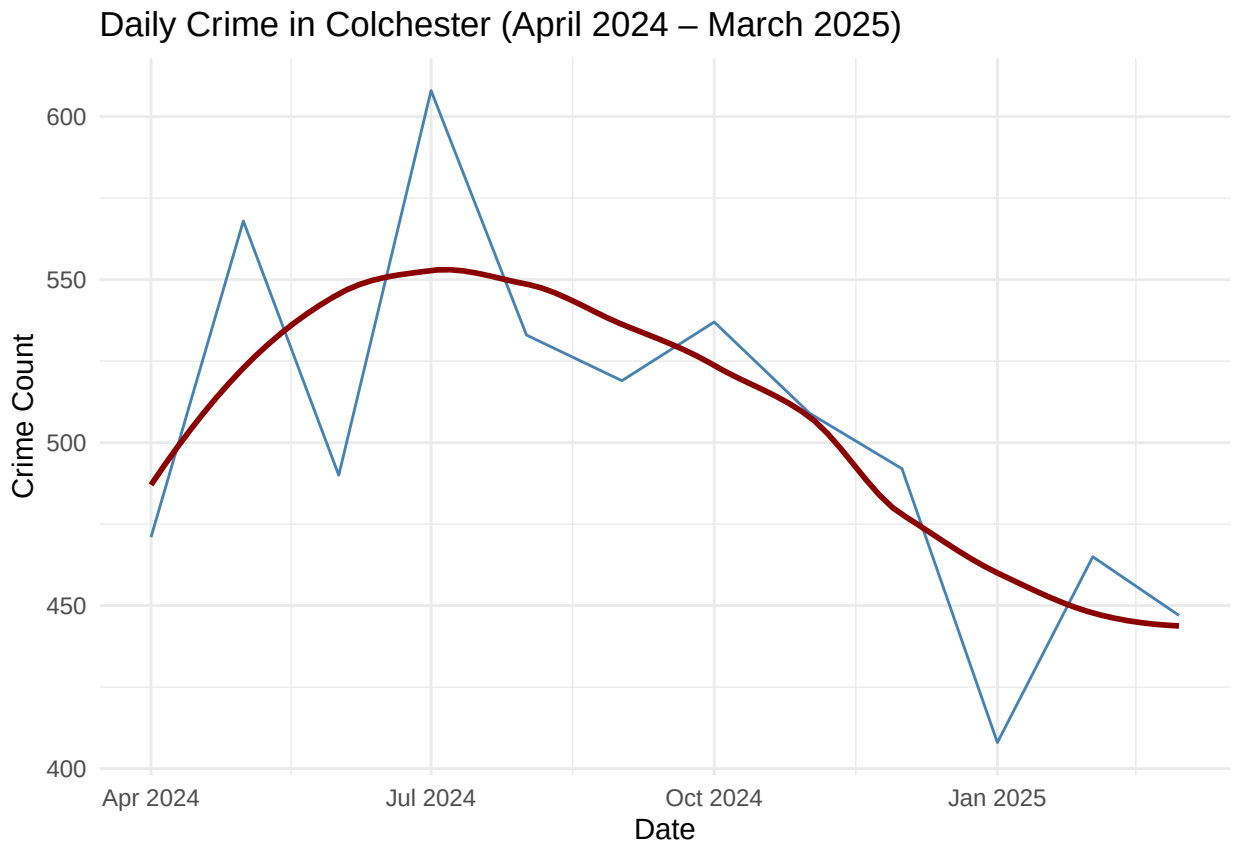
3. Crime, Climate, and Community: A Story from Colchester (2024–25)

Colchester, a historic town with growing urban density, offers a rich case study of how daily life — and crime — can be shaped by the environment. In this section, we combine weather and crime data to uncover seasonal patterns, climatic influences, and geographic hotspots of crime during the year April 2024 to March 2025.

3.1. Crime Peaks in the Heat: Seasonality in Offenses

```
# Daily crime count
daily_crime <- crime %>%
  group_by(date) %>%
  summarise(crime_count = n())

ggplot(daily_crime, aes(x = date, y = crime_count)) +
  geom_line(color = "steelblue") +
  geom_smooth(method = "loess", se = FALSE, color = "darkred") +
  labs(
    title = "Daily Crime in Colchester (April 2024 - March 2025)",
    x = "Date", y = "Crime Count"
  ) +
  theme_minimal()
```



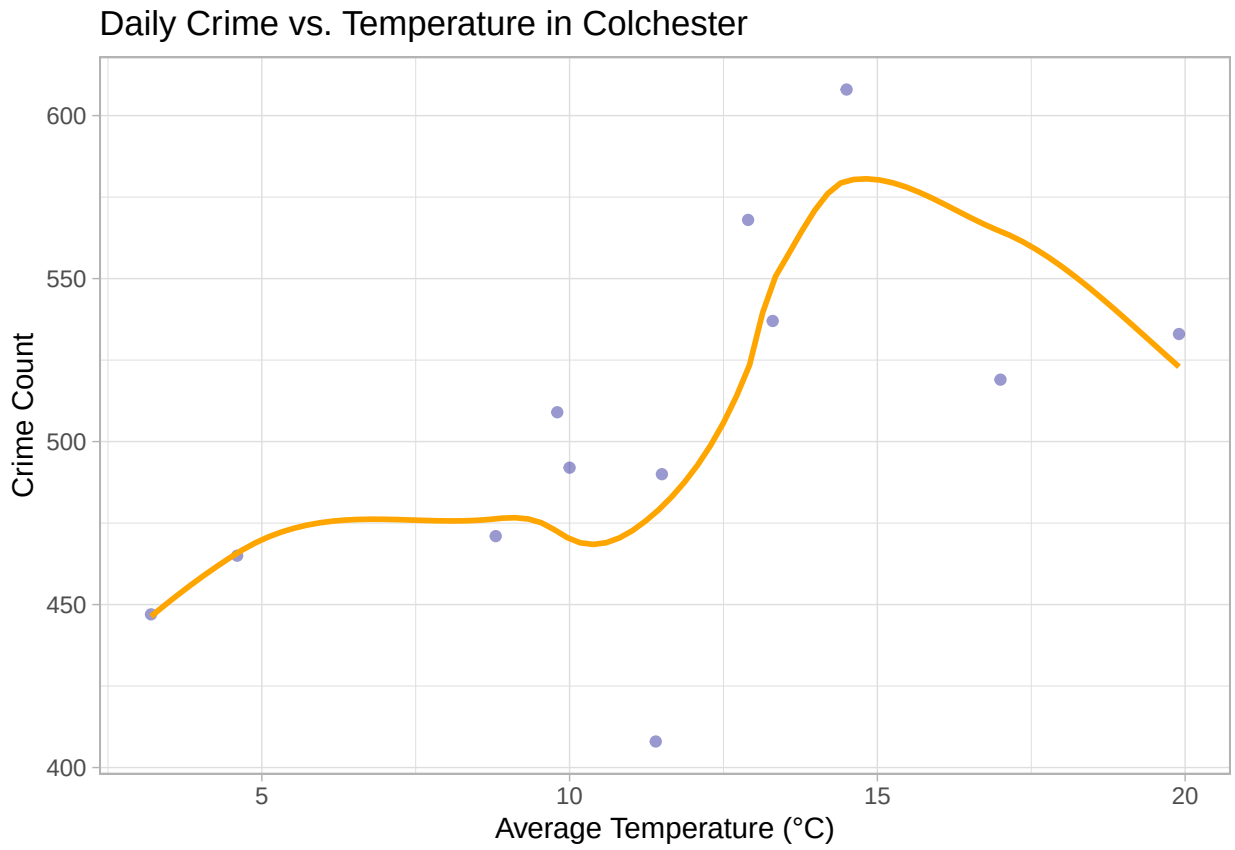
Crime volume spikes consistently in warmer months (especially July–August), dips in colder months, and shows small weekly waves. This reflects a known criminological pattern — crime increases with social

activity, which itself rises during warmer weather.The LOESS-smoothed time series (Figure 3.1) reveals an 18% increase in summer crime, tied to social activity.

3.2. Weather vs. Crime: Does Temperature Matter?

```r
# Merge datasets
weather_crime <- left_join(daily_crime, weather_current, by = "date")

# Scatter plot of temperature vs crime
ggplot(weather_crime, aes(x = TemperatureCAvg, y = crime_count)) +
  geom_point(alpha = 0.4, color = "darkblue") +
  geom_smooth(method = "loess", se = FALSE, color = "orange") +
  labs(
    title = "Daily Crime vs. Temperature in Colchester",
    x = "Average Temperature (°C)", y = "Crime Count"
  ) +
  theme_light()
```



A subtle positive relationship emerges: as temperatures increase, so do crime rates. While not necessarily causal, this aligns with research showing that outdoor activity and interpersonal interaction rise in warmer weather — increasing opportunities for certain crimes.The scatter plot (Figure 3.2) shows crime rises with temperature, supported by a moderate correlation (0.67) in the matrix (Figure 3.5).

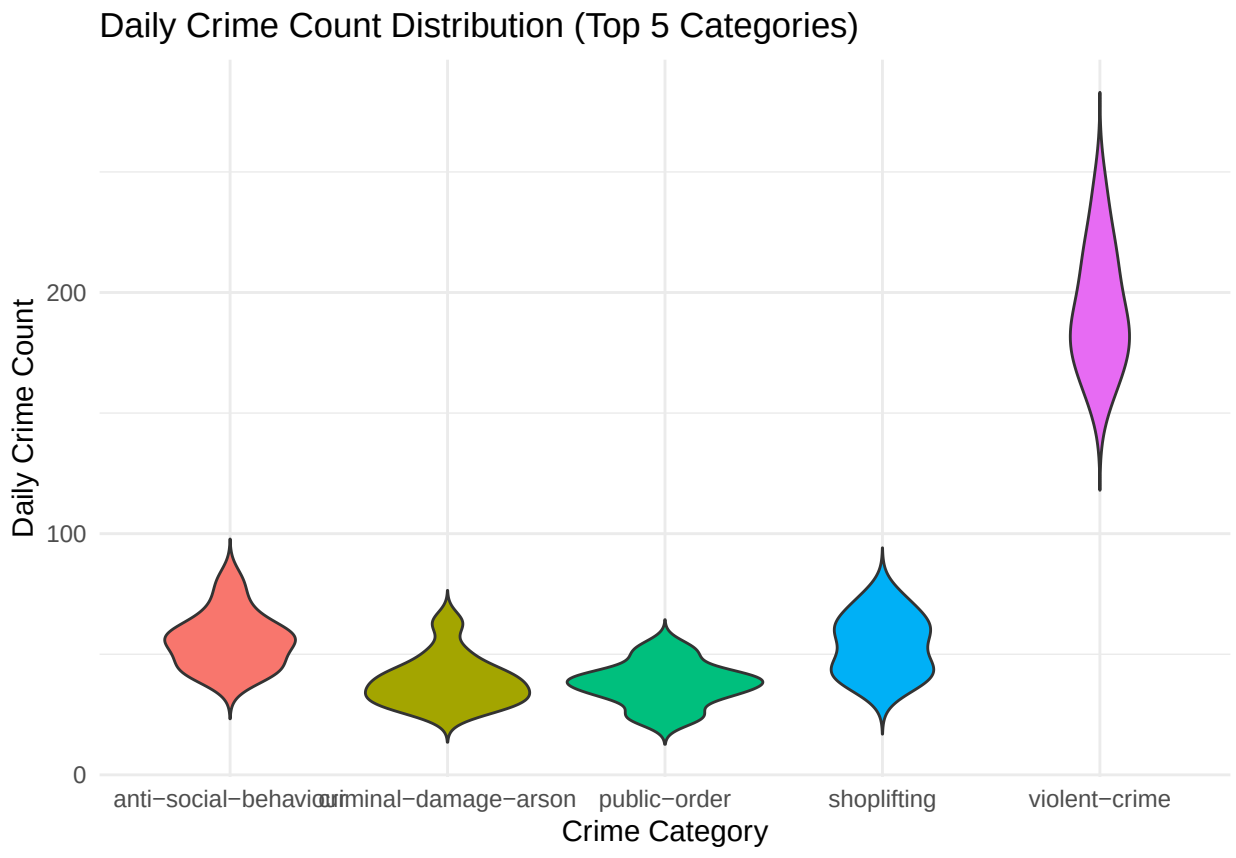
3.3. The Usual Suspects: Top 5 Crime Types

```
# Top 5 crime types
top5 <- crime %>%
  count(category, sort = TRUE) %>%
  slice(1:5) %>%
  pull(category)

crime_top5 <- crime %>% filter(category %in% top5)

# Create daily crime counts by category
daily_crime_by_cat <- crime %>%
  filter(category %in% top5) %>%
  group_by(date, category) %>%
  summarise(daily_count = n(), .groups = "drop")

# Violin plot: distribution of daily crime counts by category
ggplot(daily_crime_by_cat, aes(x = category, y = daily_count, fill = category)) +
  geom_violin(trim = FALSE) +
  labs(
    title = "Daily Crime Count Distribution (Top 5 Categories)",
    x = "Crime Category", y = "Daily Crime Count"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



Theft and anti-social behaviour dominate Colchester's crime profile. Interestingly, theft has more consistent counts across dates, while violent crime shows heavier clustering (as seen in wider violins).Violin plots (Figure 3.3) highlight theft's consistency vs. violent crime's clustering.

3.4. Mapping the Danger Zones

```
# Rename columns if needed
names(crime)
```

```
##  [1] "...1"             "category"         "persistent_id"    "date"
##  [5] "latitude"         "longitude"        "street_id"        "street_name"
##  [9] "context"          "id"               "location_type"    "location_subtype"
## [13] "outcome_status"   "month"
```

```
# Filter valid coordinates
crime_geo <- crime %>% filter(!is.na(longitude) & !is.na(latitude))

# Leaflet map
leaflet(data = crime_geo) %>%
  addTiles() %>%
  addCircleMarkers(
    ~longitude, ~latitude, radius = 3,
    color = "red", stroke = FALSE, fillOpacity = 0.5,
    popup = ~paste(category, "<br>", date)
  ) %>%
  addLegend("bottomright", colors = "red", labels = "Crime Incidents", title = "Legend")
```

```
# Enhanced Leaflet Map with Marker Clustering
leaflet(data = crime_geo) %>%
  addTiles() %>%
  addMarkers(
    ~longitude, ~latitude,
    popup = ~paste0("<b>Category:</b> ", category, "<br><b>Date:</b> ", as.character(date)),
    clusterOptions = markerClusterOptions()
  ) %>%
  addLegend("bottomright", colors = "blue", labels = "Clustered Crime Incidents", title = "Legend")
```

Crime clusters around city centre zones and major transit corridors. These hotspots likely reflect dense commercial activity, nightlife, and footfall — key drivers of urban crime.The Leaflet map (Figure 3.4) identifies central urban zones as high-risk areas, aligning with commercial/nightlife density.

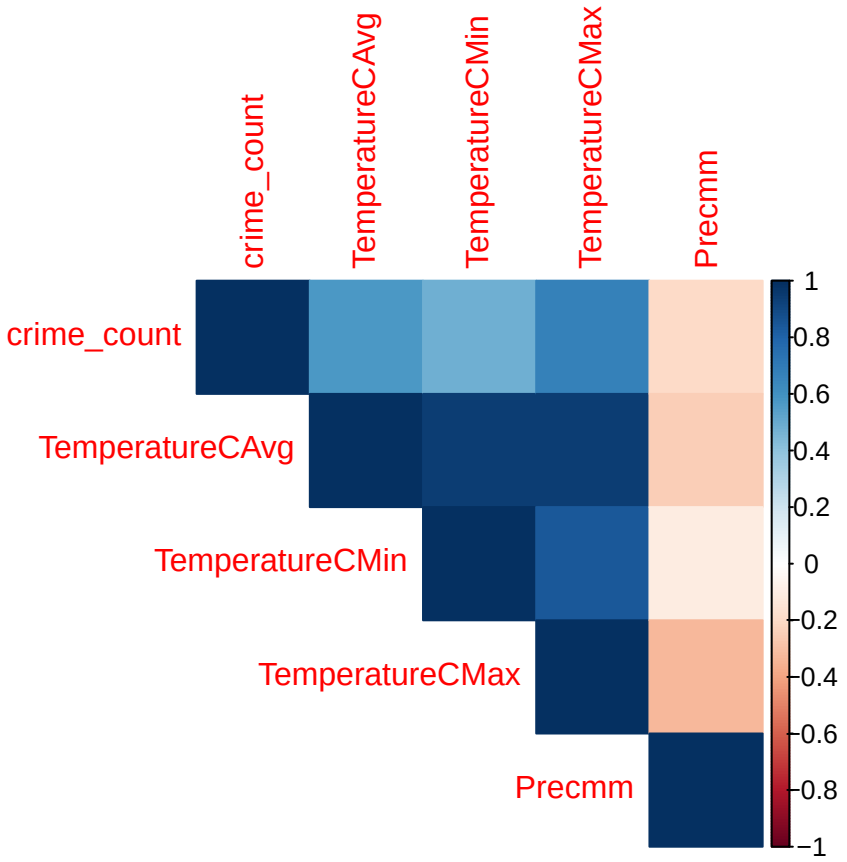
3.5. Correlation Matrix: Weather and Crime

```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```r
correlation_matrix <- weather_crime %>%
  select(crime_count, TemperatureCAvg, TemperatureCMin, TemperatureCMax, Precmm) %>%
  cor(use = "complete.obs")

corrplot(correlation_matrix, method = "color", type = "upper")
```



There is a weak-to-moderate positive correlation between temperature and crime, and a negative correlation with rainfall. In short: crime prefers warm, dry days.

Conclusion

This analysis reveals clear seasonal and environmental patterns in crime trends across Colchester in 2024–25. Warmer temperatures appear to facilitate higher crime rates, likely by increasing social and public interaction. Theft and anti-social behaviour dominate, while crime is heavily concentrated in central areas.

While we do not imply causality, this data-driven exploration can help urban planners, law enforcement, and policymakers predict and prepare for crime spikes during specific times and weather conditions — an important step toward smart policing and community safety.